

AETHERIA // LIFE RPG

Official User Manual & Operator Handbook

VERSION 1.0 • HACKATHON EDITION • FULLSTACK 3D PROGRESSION

1. THE CORE PHILOSOPHY

Bridging the Delayed Gratification Gap

Traditional productivity applications (to-do lists, reminder apps, and habit trackers) fail because they treat life like a series of administrative chores. They suffer from the delayed gratification problem: studying for 2 hours or going to the gym does not yield visible real-world changes for weeks. Video games, however, deliver immediate dopamine via tactile feedback, clear progression ladders, and visible trophies.

Aetheria converts your everyday habits, work tasks, and fitness routines into an engaging 3D Cyber-Fantasy Role-Playing Game. Completing real tasks earns Experience Points (XP), in-game Gold, and Attribute Boosts, while dealing direct damage to raid boss monsters.

How to Access & Run the Platform

The application runs completely locally with zero external configuration or cloud lock-in:

- **1. Open Terminal:** Navigate to C:\Users\de\l\Desktop\life-rpg
- **2. Launch App:** Execute: npm run dev
- **3. Open Browser:** Visit http://localhost:3000 in Chrome, Edge, or Firefox

& 1-Click Judge Demo Login (Zero-Setup Evaluation)

Judges can skip manual registration by clicking '1-CLICK DEMO CHARACTER LOGIN' on the landing screen. You will immediately enter as CyberKnight_Alex (Level 3 Aether Initiate) with preloaded quests, an equipped Cyber Katana, 318 Gold, and an active Boss Raid.

System Architecture At A Glance

- **Frontend:** Next.js 14 App Router, React 18, Tailwind CSS, Lucide Icons, Canvas-Confetti
- **3D Engine:** Three.js WebGL procedural avatar with dynamic equipment binding
- **Sound Engine:** Synthesized Web Audio API (Zero external MP3 dependency, 100% reliable)
- **Backend API:** Server-authoritative Route Handlers (/api/*) with anti-cheat verification
- **Database:** Prisma ORM with SQLite (local) + 1-click PostgreSQL support for production

2. INTERACTIVE 3D HERO AVATAR

Visual Progression & Dynamic Equipment

At the center of your screen is your personal 3D Hero Avatar rendered via WebGL Three.js. This is not a static 2D image; it is an interactive 3D procedural character.

- **360-Degree Mouse Drag:** Click and drag anywhere on the 3D canvas with your mouse or finger to rotate and inspect your hero from all angles.
- **Idle Bobbing & Aura:** Your character continuously hovers with subtle breathing animations and an orbiting stardust particle field.
- **Dynamic 3D Equipment Swapping:** Whenever you purchase or equip gear in the Relic Armory, the 3D model immediately updates in real time:
 - Weapon (Cyber Katana / Plasma Blade): Attaches a glowing laser sword to the right arm.
 - Shield (Quantum Hex Shield / Void Aegis): Attaches a translucent energy barrier to the left arm.
 - Wings (Neon Photon Wings / Archangel Wings): Unfurls holographic angular cyber-wings on the back.
 - Crown (Neural Focus Visor / Sovereign Crown): Emits a glowing halo crown above the head.

3. QUESTS & DAILY RITUALS

How to Organize and Complete Real-World Tasks

Quests represent your real-life productivity goals. Click the 'Quests & Daily Rituals' tab to manage them.

Quest Types Explained

- **Daily Rituals (DAILY):** Recurring daily commitments (e.g., 'Morning Workout', 'Study Next.js'). Reset daily to build rock-solid routines.
- **Repeatable Habits (HABIT):** Habits you can complete multiple times a day (e.g., 'Drink 500ml Water', '15-min Meditation').
- **One-Time Bounties (BOUNTY):** Single-instance tasks with large payouts (e.g., 'Clean Desk & Cable Management', 'Submit Hackathon Project').

How to Forge a New Quest

- **Step 1:** Click the '+ NEW QUEST' button (or press the 'N' shortcut key on your keyboard).
- **Step 2:** Enter Quest Title (e.g. 'Solve 2 LeetCode Mediums') and optional description notes.
- **Step 3:** Select Type (Daily, Habit, Bounty) and Target Attribute (Intellect, Strength, Agility, Vitality, Spirit).
- **Step 4:** Choose Difficulty (Trivial, Easy, Medium, Hard, Heroic). Notice the Guaranteed Loot Preview updating dynamically!
- **Step 5:** Click 'FORGE QUEST'. It instantly syncs with the database.

🏆 Completing a Quest & Micro-Interactions

Click the circular checkbox next to any quest. Instantly, the Web Audio engine sounds a chime, a burst of celebratory particle confetti fires, the quest marks complete, XP and Gold are added, and kinetic strike damage is dealt to the Boss!

4. THE RPG PROGRESSION ENGINE

Non-Linear Mathematics & Server Anti-Cheat

Aetheria implements a mathematically rigorous, server-authoritative non-linear progression engine. Users cannot spoof stats via client-side inspect element.

Non-Linear Leveling Curve

Each subsequent level demands exponentially more dedication than the previous rank:

XP Formula: $XP_Required(Level) = \text{Math.floor}(100 * (Level ^ 1.55))$

Level 1 -> 2: 100 XP | Level 2 -> 3: 292 XP | Level 5 -> 6: 1,211 XP | Level 10 -> 11: 3,548 XP

Level Up Celebration & Stat Allocation

When your XP crosses the threshold, a celebratory Level Up Modal emerges with dual-angle confetti fireworks and victory fanfare. Each level up awards +2 Unspent Stat Points to distribute into your Sovereign Attribute Matrix.

5. BOSS BATTLE RAID ARENA

Collaborative Productivity Combat

Your procrastination and digital distractions are personified as an active Raid Boss: 'Malakor the Procrastinator' (Lord of Endless Delay).

- **Dealing Damage:** Completing tasks converts your gained XP into kinetic boss damage (Damage = 85% of total XP).
- **Floating Damage Popups:** Attacking triggers visual damage flashes ('-45 HP!') and a cyber strike sound.
- **Boss Slaying Bounty:** Depleting the Boss HP to 0 slays the beast, awarding a +150g Victory Bounty and spawning the next higher tier Boss!

6. RELIC ARMORY & IN-GAME ECONOMY

Spending Your Gold on Tangible Upgrades

- **Browse Relics:** Weapons, Shields, Wings, Helms, and Potions across Common, Rare, Epic, and Legendary tiers.
- **Purchase Gear:** Gold earned from completing quests can be spent in the shop. Server verifies balance before granting items.
- **Equip & Unequip:** Equipping an item immediately applies its stat bonuses and renders it onto the 3D Hero Avatar!

7. THE SOVEREIGN ATTRIBUTE MATRIX

5-Dimensional Life Balancing

Tasks level up specific dimensions of your real-world capability, mapped visually in the interactive SVG Radar Pentagonagram:

- **&"p Strength (Gym & Bore)** Raised via workouts, calisthenics, physical activity. Multiplies boss strike damage.
- **Ø>Ýà Intellect (Study & Code)** Raised via programming, reading, problem solving. Boosts base XP reward rates.
- **&i Agility (Speed & Execution)** Raised via quick errands, inbox zero, speed cleaning. Multiplies daily streak bonuses.
- **Ø<ß? Vitality (Health & Sleep)** Raised via drinking 2.5L water, 8h quality sleep, nutrition. Expands maximum HP pool.
- **Ø=Ý. Spirit (Mindfulness)** Raised via meditation, deep breathing, journaling. Fortifies mental resilience.

8. DATABASE PERSISTENCE & AUDIT CHRONICLES

Proving Real Data Integrity

To satisfy the hackathon's strict Zero-Tolerance rule against fake localStorage persistence, Aetheria persists all state in a relational database via Prisma ORM.

- **Historical Audit Chronicles:** Click the 'Historical Audit Chronicles' tab to inspect every quest completed, with exact timestamps, XP, and gold earned stored permanently in the database.
- **F5 Page Refresh Test:** Press F5 or reload the browser at any time. All user levels, equipment, equipped 3D meshes, quests, and audit logs remain completely persistent!

9. KEYBOARD SHORTCUTS & ACCESSIBILITY

Fluid Power-User Navigation

- **Tab / Shift+Tab:** Navigate seamlessly between quest cards, buttons, and filters.
- **Space / Enter:** Complete the currently focused quest with instant keyboard activation.
- **Mute / Unmute Button:** Located in the top header to toggle synthesized sound effects on or off.

10. TROUBLESHOOTING & FAQ

- **Q: Sound is not playing?:** Click the sound icon in the header navigation or click anywhere on the page to wake the browser AudioContext.
- **Q: How do I test level up quickly?:** Complete the 'Master Next.js Fullstack Architecture' quest (Hard difficulty) or forge a Heroic quest for 160 XP!
- **Q: Where is the database file stored?:** Inside C:\Users\del\l\Desktop\life-rpg\dev.db.